Case Study: Retail Sales Insights Dashboard

Latreche Sara - code213

Project Context and Objective

Imagine you've just joined a retail company as a **Data Analyst**. On your first day, your manager gives you your first assignment:
"We need a **Sales Insights Dashboard** — something clear, automated, and interactive. It should help us track performance across products, regions, and sales reps. You'll have access to all the raw data — just make sense of it and make it easy for everyone to use."
This is both exciting and challenging — a chance to build something from scratch that could directly support decision-making at the company.

Step 1: Understand the Requirements

- Define the goal: build a **one-page Power BI dashboard** to display sales insights across multiple dimensions — product, geography, and time.
- Identify and gather the required data sources:
- Sales data (stored by year in a folder)
- Categories (Excel file)
- SubCategories (Excel file)
- Sales Representatives (Excel file)
- Product data (CSV or database)
- Geography data (Excel file)
- Ensure the system design is **resilient**:
- Removing a yearly sales file should not cause refresh errors.
- Adding a new sales file should automatically include it in the dataset upon refresh.

Step 2: Data Preparation

•Task 1.1 - Automate Sales Data Loading

•Use **Power Query** to create a **dynamic loading mechanism** that imports all yearly sales files from the sales folder.

•Combine all files into a single **Sales Fact Table**.

•Ensure the model automatically updates when new files are added, without requiring manual intervention.

Step 3: Data Cleaning and Transformation

•Task 1.1 – Automate Sales Data Loading

•Task 2.1 – Split and Format Fields

•Split the **Location** field in the Sales table into two separate columns: *City* and *Country*.

•Assign appropriate **data types** for use in Power BI maps.

•Standardize the **Date field** to support time intelligence calculations in DAX.

•Task 2.2 – Create GeoKey

•Create a unique **GeoKey** column in both the **Sales** and **Geography** tables.

•Use this key to establish relationships between sales transactions and geographic locations.

•Task 2.3 – Clean ID Columns

•Identify columns in the **SalesRep** and **SubCategory** tables that contain IDs with the prefix “ID - ”.

•Build a reusable **Power Query function** that removes this prefix.

•Apply the function to both tables to clean and standardize the ID columns.

•Task 2.4 – Build the Data Model

•Design a **star schema** model:

- •**Fact Table:** Sales
- •**Dimension Tables:** Product, Category, SubCategory, SalesRep, Geography, Calendar

•Validate relationships between tables to ensure accurate aggregation and reporting.

Step 4: Create Business Measures with DAX

•Define DAX measures to calculate key performance metrics:
•**Total Revenue** = Units × Product Retail Price
•**Total Cost** = Units × Product Standard Cost
•**Gross Profit** = Total Revenue – Total Cost
•**Gross Profit MoM Growth %** = Month-over-Month change in Gross Profit
•**Average Sales per Day** = Total Revenue ÷ Number of Active Sales Days
•**QoQ Growth** = Quarter-over-Quarter growth (for QBR reporting)
•Verify all measures for accuracy and consistency using sample data.

Step 5: Design the Dashboard

•Use Power BI visuals to create an **interactive one-page dashboard**.
•Include the following elements:
•**KPI Cards:** Total Revenue, Total Cost, Gross Profit
•**Line Charts:** Month-over-Month (MoM) and Quarter-over-Quarter (QoQ) growth
•**Maps:** Sales distribution by Country and City
•**Bar Charts:** Top Products and Sales Representatives
•**Filters/Slicers:** Year, Country, Category
•Ensure months are sorted chronologically from **January to December**.
•Design the layout for clarity, readability, and executive-level presentation.

and Share the Report

Step 6: Deploy and Share the Report

•Publish the Power BI report following these deployment steps:

•Create a **Power BI Workspace** and assign appropriate user permissions.

•**Publish** the report to the Power BI Service.

•Configure a **Gateway** to enable on-premises data refresh.

•Set up **Scheduled Refreshes** for automated updates.

•Add **Roles, Subscriptions, and Alerts** for stakeholders.

•Package the report into a **Power BI App** and share it across departments.

and Share the Report

Step 7: Evaluate Results

•Verify that the dashboard:
•Updates automatically when new yearly data is added.
•Provides accurate and real-time performance insights.
•Enables managers to analyze sales trends by product, region, and representative.
•Confirm that the dashboard supports **data-driven decisions** during quarterly business reviews (QBRs).
•Document feedback from management and optimize visuals or metrics if needed.